

Programme Code(s): TU856, TU857

Module Code(s): CMPU 3007

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TECHNOLOGICAL UNIVERSITY DUBLIN
CITY CAMPUS, GRANGEGORMAN

TU856 – BSc (Hons) in Computer Science
TU857 – BSc (Hons) in Computer Science (Infrastructure)

Year 3

SEMESTER 1 EXAMINATIONS 2025/26

CMPU 3007 Cloud Computing

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Instructions To Candidates: The examination paper contains FOUR questions.
You must answer question 1 and any TWO questions from the remaining three questions.
Please start each question on a new page.

Exam Duration: 2 hours

Special Instructions /Handouts/ Materials Required: NA

Question 1 – (Compulsory) [40 Marks]

Write a detailed explanatory note on all **FOUR (4)** of the following topics:
(10 marks each)

- (a) Essential Characteristics of Cloud Computing (according to NIST).
- (b) The Role and Functionality of a Hypervisor.
- (c) Benefits and Limitations of Virtual Machine Snapshots.
- (d) Defence-in-Depth.

[Question One Total Marks 40]

Answer any TWO from Questions 2, 3 & 4

Question 2 [30 Marks]

- (a) Discuss and critically assess the following **THREE (3)** best practices of the Twelve Factor App from the point of view of a distributed Cloud native application.

- i. *Concurrency* (5 marks)
- ii. *Disposability* (5 marks)
- iii. *Dev/Prod Parity* (5 marks)

- (b) A media-streaming company expects sudden spikes in user traffic during major sports events and wants to minimise infrastructure overhead when demand is low.

- i. Explain how a serverless architecture can support the company's *scalability* and *cost-efficiency* objectives (5 marks)
- ii. Describe how a serverless approach can improve the *reliability* and *responsiveness* of streaming services delivered to a global audience. (5 marks)
- iii. Identify **TWO (2)** potential risks or limitations of using serverless in this context and outline appropriate mitigation strategies. (5 marks)

[Question Two Total Marks 30]

Question 3 [30 Marks]

- (a) When evaluating potential Cloud Service Providers, outline the key questions and considerations a medium-to-large organisation should address before deciding which provider to engage. **(15 marks)**
- (b) A software team wants to migrate its applications from Virtual Machines to Docker Containers to improve efficiency.
- i. Describe what Docker Containerisation is. **(5 marks)**
 - ii. Contrast its architectural approach with that of Virtual Machines **(6 marks)**
 - iii. Briefly discuss the features that make Docker more lightweight and scalable. **(4 marks)**

[Question Three Total Marks 30]

Question 4 [30 Marks]

- (a) A software team plans to introduce Continuous Integration (CI) as part of its DevOps transformation.
- i. Outline what the CI process involves. **(5 marks)**
 - ii. Evaluate the *team responsibilities* and *practices* required to achieve consistent, stable builds. **(5 x 2 = 10 marks)**
- (b) Continuous Delivery (CD) enables mid-to-large organisations to deploy software safely, quickly, and routinely. Critically evaluate the impact of Continuous Delivery by discussing **FIVE (5)** key benefits it provides to such organisations. **(3 x 5 = 15 marks)**

[Question Four Total Marks 30]

<END OF EXAMINATION PAPER>